# Appendix: Verified-Failure Cost-Aware Safe Stopping

A high-probability guarantee for adaptive failure-driven acquisition

**Purpose.** This appendix formalizes one theorem for the proposed method. The guarantee is intentionally different from ACT's parameter-gap bound: it bounds the *best remaining true cost-adjusted gain* among robustly verified failure modes at the adaptive stopping round. The presentation follows the theorem–proof style used in ACT's theoretical appendix, but the result below is a proposed theorem, not a theorem from ACT.

## A.1 Mathematical flow

At active-learning round $t$, the current model is $\theta_t$ and the complete history available before the next decision is $\mathcal{F}_t$. The method proceeds as

$$\boxed{\text{errors}} \longrightarrow \boxed{\mathcal{H}_t\text{: failure hypotheses}} \longrightarrow \boxed{\text{verification probes}}$$

$$\longrightarrow \boxed{\widehat{\mathcal{V}}_t\text{: certified failures}} \longrightarrow \boxed{\text{gain UCB} - \lambda\,\text{cost}} \longrightarrow \boxed{\text{fix or stop}}.$$

For each candidate $h \in \mathcal{H}_t$, verification probes produce Bernoulli outcomes

$$Z_{t,j}(h) \in \{0,1\}, \qquad \nu_t(h) := \mathbb{E}[Z_{t,j}(h) \mid \mathcal{F}_t, h],$$

where $Z_{t,j}(h) = 1$ means that the model behavior on probe $j$ is consistent with the failure pattern predicted by hypothesis $h$. With $n_t(h)$ probes,

$$\widehat{\nu}_t(h) = \frac{1}{n_t(h)} \sum_{j=1}^{n_t(h)} Z_{t,j}(h), \qquad a_t(h) = \sqrt{\frac{\log(4KT/\delta)}{2n_t(h)}}.$$

A hypothesis is *certified* when its lower confidence bound exceeds the verification threshold $\tau_v$:

$$\boxed{\widehat{\mathcal{V}}_t := \{h \in \mathcal{H}_t : \widehat{\nu}_t(h) - a_t(h) \geq \tau_v\}.} \tag{A.1}$$

For each certified $h$, let the true expected downstream gain of a targeted intervention be

$$\boxed{\Delta_t(h) := \mathbb{E}\left[R(\theta_t) - R\left(\theta_{t+1}^{(h)}\right) \mid \mathcal{F}_t, h\right],} \tag{A.2}$$

where $R(\theta)$ is validation risk and $\theta_{t+1}^{(h)}$ denotes the model after acquiring targeted data for $h$ and updating the model.

Assume the method obtains $m_t(h)$ fresh pilot gain observations $X_{t,j}(h) \in [-M, M]$ satisfying

$$\mathbb{E}[X_{t,j}(h) \mid \mathcal{F}_t, h] = \Delta_t(h).$$

Define

$$\widehat{\Delta}_t(h) = \frac{1}{m_t(h)} \sum_{j=1}^{m_t(h)} X_{t,j}(h), \qquad b_t(h) = M\sqrt{\frac{2\log(2KT/\delta)}{m_t(h)}}. \tag{A.3}$$

Let $C_t(h) \geq 0$ be the known total cost of fixing $h$ at round $t$, and let $\lambda > 0$ convert cost into the same decision scale as gain. The true net value and its optimistic upper bound are

$$J_t(h) = \Delta_t(h) - \lambda C_t(h), \qquad \boxed{U_t(h) = \widehat{\Delta}_t(h) + b_t(h) - \lambda C_t(h).} \tag{A.4}$$

The algorithm continues with the certified failure having largest $U_t(h)$ if that maximum is positive; otherwise it stops:

$$\boxed{\tau := \inf\left\{t \leq T : \max_{h \in \widehat{\mathcal{V}}_t} U_t(h) \leq 0\right\}.} \tag{A.5}$$

We use the convention $\max \emptyset = -\infty$.

## A.2 Variables and assumptions

| Symbol | Meaning |
|---|---|
| $t$, $T$ | Current round and a finite maximum number of rounds. |
| $\mathcal{F}_t$ | All information observed before the decision at round $t$. |
| $\mathcal{H}_t$ | Candidate failure hypotheses at round $t$; assume $\lvert\mathcal{H}_t\rvert \le K$. |
| $Z_{t,j}(h)$ | Binary verification outcome for probe $j$ of hypothesis $h$. |
| $\nu_t(h)$ | True probability that probes exhibit the behavior predicted by $h$. |
| $\tau_v$ | Verification threshold. A larger value requires stronger reproducibility. |
| $\gamma$ | Verification margin separating clearly valid and clearly invalid hypotheses. |
| $\widehat{\mathcal{V}}_t$ | Failures certified by a lower confidence bound, Eq. (A.1). |
| $R(\theta)$ | Validation risk/loss of model parameters $\theta$; smaller is better. |
| $\Delta_t(h)$ | True expected risk reduction obtained by fixing failure $h$. |
| $X_{t,j}(h)$ | Fresh bounded pilot observation used to estimate $\Delta_t(h)$. |
| $M$ | Known bound such that $X_{t,j}(h) \in [-M, M]$. |
| $C_t(h)$ | Total intervention cost: human review + LLM + data acquisition + training/evaluation. |
| $\lambda$ | Cost-to-gain trade-off coefficient. |
| $J_t(h)$ | True cost-adjusted value, $\Delta_t(h) - \lambda C_t(h)$. |
| $U_t(h)$ | Upper confidence bound on $J_t(h)$. |
| $\delta$ | Allowed probability that the simultaneous guarantee fails. |
| $\tau$ | Adaptive stopping round from Eq. (A.5). |

**Assumption 1** (Conditional concentration and verification margin)**.** *For every evaluated pair $(t, h)$:*

1. *conditioned on $(\mathcal{F}_t, h)$, the verification probes $Z_{t,j}(h)$ are independent Bernoulli variables;*
2. *conditioned on $(\mathcal{F}_t, h)$, the pilot gain observations $X_{t,j}(h)$ are independent, satisfy $X_{t,j}(h) \in [-M, M]$, and have mean $\Delta_t(h)$;*
3. *every candidate is separated from the verification boundary by a margin $\gamma > 0$, i.e. either $\nu_t(h) \ge \tau_v + \gamma$ or $\nu_t(h) \le \tau_v - \gamma$;*
4. *the number of verification probes is large enough that $a_t(h) \le \gamma/2$;*
5. *$C_t(h)$ is $\mathcal{F}_t$-measurable (known before the continue/stop decision).*

## A.3 Main theorem

Define the set of *robustly valid* failures at round $t$ as

$$\mathcal{V}_t^{\gamma} := \{h \in \mathcal{H}_t : \nu_t(h) \geq \tau_v + \gamma\}. \tag{A.6}$$

**Theorem 1** (Verified-Failure Cost-Aware Safe Stopping)**.** *Suppose Assumption 1 holds, $|\mathcal{H}_t| \leq K$ for all $t \leq T$, and the algorithm uses Eqs. (A.1)–(A.5). Then, with probability at least $1-\delta$, at the adaptive stopping round $\tau$,*

$$\boxed{\max_{h \in \mathcal{V}_\tau^{\gamma}} [\Delta_\tau(h) - \lambda C_\tau(h)] \leq 0.} \tag{A.7}$$

*Equivalently, with probability at least $1-\delta$, when the method stops there is no robustly verified failure mode whose true expected downstream gain exceeds its cost-adjusted intervention cost.*

*Proof.* The proof has three steps.

**Step 1: simultaneous correctness of failure certification.** For a fixed evaluated pair $(t, h)$, Hoeffding's inequality for Bernoulli observations gives
Here, the empirical estimator is the average: sum the n_t(h) Bernoulli probes and divide by n_t(h).

$$\mathbb{P}(|\widehat{\nu}_t(h) - \nu_t(h)| > a_t(h) \mid \mathcal{F}_t, h) \leq 2\exp\big(-2n_t(h)a_t(h)^2\big) = \frac{\delta}{2KT}. \tag{A.8}$$

There are at most $KT$ evaluated round–hypothesis pairs. A union bound therefore yields the event

$$\mathcal{E}_v := \{|\widehat{\nu}_t(h) - \nu_t(h)| \leq a_t(h),\ \forall (t,h)\}$$

with

$$\mathbb{P}(\mathcal{E}_v) \geq 1 - \frac{\delta}{2}. \tag{A.9}$$

On $\mathcal{E}_v$, take any robustly valid $h \in \mathcal{V}_t^{\gamma}$. Since $\nu_t(h) \geq \tau_v + \gamma$ and $a_t(h) \leq \gamma/2$,

$$\widehat{\nu}_t(h) - a_t(h) \geq \nu_t(h) - 2a_t(h) \geq \tau_v + \gamma - 2a_t(h) \geq \tau_v. \tag{A.10}$$

Hence every robustly valid failure is certified:

$$\boxed{\mathcal{V}_t^{\gamma} \subseteq \widehat{\mathcal{V}}_t \quad \forall t} \qquad \text{on } \mathcal{E}_v. \tag{A.11}$$

**Step 2: simultaneous upper confidence bound on intervention gain.** For a fixed evaluated pair $(t, h)$, the pilot observations lie in $[-M, M]$. The one-sided Hoeffding inequality gives

$$\mathbb{P}\Big(\Delta_t(h) > \widehat{\Delta}_t(h) + b_t(h) \mid \mathcal{F}_t, h\Big) \leq \exp\left(-\frac{m_t(h)b_t(h)^2}{2M^2}\right) = \frac{\delta}{2KT}. \tag{A.12}$$

Another union bound over at most $KT$ evaluated pairs gives the event

$$\mathcal{E}_g := \Big\{\Delta_t(h) \leq \widehat{\Delta}_t(h) + b_t(h),\ \forall (t,h)\Big\}$$

with

$$\mathbb{P}(\mathcal{E}_g) \geq 1 - \frac{\delta}{2}. \tag{A.13}$$

Therefore, on $\mathcal{E}_g$,

$$J_t(h) = \Delta_t(h) - \lambda C_t(h) \leq \widehat{\Delta}_t(h) + b_t(h) - \lambda C_t(h) = U_t(h) \tag{A.14}$$

for every evaluated $(t, h)$.

**Step 3: apply the uniform bounds at the adaptive stopping round.** By the union bound,

$$\mathbb{P}(\mathcal{E}_v \cap \mathcal{E}_g) \geq 1 - \delta. \tag{A.15}$$

Crucially, both events hold *simultaneously for every round and every evaluated hypothesis*; therefore they remain valid at the random, data-dependent stopping time $\tau$.

On $\mathcal{E}_v \cap \mathcal{E}_g$, Eq. (A.11) implies

$$\mathcal{V}_\tau^{\gamma} \subseteq \widehat{\mathcal{V}}_\tau. \tag{A.16}$$

For any $h \in \mathcal{V}_\tau^\gamma$, Eq. (A.14) gives $J_\tau(h) \le U_\tau(h)$. Hence

$$\max_{h \in \mathcal{V}_\tau^\gamma} J_\tau(h) \le \max_{h \in \widehat{\mathcal{V}}_\tau} U_\tau(h). \tag{A.17}$$

By definition of the stopping rule in Eq. (A.5),

$$\max_{h \in \widehat{\mathcal{V}}_\tau} U_\tau(h) \le 0. \tag{A.18}$$

Combining Eqs. (A.17) and (A.18),

$$\max_{h \in \mathcal{V}_\tau^\gamma} [\Delta_\tau(h) - \lambda C_\tau(h)] \le 0. \tag{A.19}$$

Since Eq. (A.19) holds on an event of probability at least $1 - \delta$, the result follows. □

## A.4 What is actually bounded?

The theorem does **not** bound a parameter distance such as $\|\theta_*^{ACT} - \theta^*\|$. Instead, it bounds the best remaining *decision value* at stopping:

$$\underbrace{\max_{h \in \mathcal{V}_\tau^\gamma} \Delta_\tau(h)}_{\text{best remaining true expected improvement}} - \underbrace{\lambda C_\tau(h)}_{\text{cost-adjusted intervention price}} \le 0.$$

Thus, under the stated assumptions, stopping is statistically safe: with probability at least $1 - \delta$, no robustly verified failure still offers positive expected net value.

**Verification checklist for a real implementation.** The theorem becomes empirically defensible only if the implementation specifies: (i) how counterfactual/invariant probes produce the Bernoulli $Z_{t,j}(h)$; (ii) a concrete pilot procedure whose observations $X_{t,j}(h)$ are approximately conditionally unbiased for the downstream gain $\Delta_t(h)$; (iii) a defensible bound $M$; and (iv) the cost model $C_t(h)$. These are assumptions to validate experimentally, not consequences of the theorem itself.